Checkpoint 4 — Synthetic Data Generator Design

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1 Overview

This checkpoint documents the full design of the synthetic data generator, covering the three-layer generation architecture, archetype behavioral priors, population proportions, code structure, output format, and the anomaly injection strategy. All design decisions are calibrated against source extractions S1–S3 and verified against external data where possible.

2 Architecture Decisions

Two architecture changes were locked during this session:

Decision	Rationale
Channel dropped from LSTM prediction heads (7 → 6)	All four caisse operations (retrait, versement, virement, remise de chèques) are teller-window (guichet) operations. The `channel` field remains in the raw event schema but carries zero entropy in the training data, making it useless as a prediction target.
Remise de chèques revised downward across all archetypes	Law n°41-2024 (effective February 2025) introduced a 30,000 DT per-cheque ceiling and mandated banks encourage alternative payment methods. BCT Q1 2026 data confirms a 24.9% volume decline in cheque usage. B2B transactions increasingly use virements and promissory notes.
New anomaly pattern (D23)	Cheque structuring near the 30,000 DT ceiling, analogous to the 10,000 DT smurfing pattern. Same detection logic via `near_threshold_ratio`.

3 Three-Layer Generation Architecture

Layer	Scope	Mechanism	Purpose
Layer 1: Archetype	Population-level	Fixed mean/std per dimension	Defines central tendency for each client type
Layer 2: Personality	Per-client (sampled once)	Gaussian draw from archetype parameters	Gives each client a stable individual baseline

Layer	Scope	Mechanism	Purpose
Layer 3: Noise	Per-event	Gaussian jitter around personality	Prevents deterministic event sequences

Standard deviations narrow progressively: archetype (widest) → personality → noise (tightest).

4 Archetype Behavioral Priors

Six behavioral dimensions define each archetype. Values represent the archetype-level central tendency (Layer 1 means); Layer 2 personality sampling varies individuals around these centers.

4.1 Monthly Operation Counts

Operation	Salaried	Small Biz	Student	Retiree	Big Biz
Retrait	2–4	4–8	∼8	1–2	0–2
Versement	∼1/quarter	20–25	∼0	∼0	8–10
Virement	1–3	7–10	∼0	1–2	30–70+
Remise de chèques	∼1/quarter	1–2	0	∼0	2–4
Total	4–8	30–40	∼8	2–4	40–80+

4.2 Behavioral Dimensions

Dimension	Salaried	Small Biz	Student	Retiree	Big Biz
Dominant op	retrait	versement	retrait	retrait	virement
Timing	Salary cycle (start of month)	Daily rhythm (morning before branch close)	Irregular	Pension cycle (start of month)	Business cycle (end-of-month payroll, supplier dates)
Amount	Stable, salary-proportional	Varied, medium	Small, irregular	Fixed, low	High, varied
Counterparty	Fixed/few	Varied	Parents/none	None/fixed	Many, varied
Branch loyalty	High	Medium	High	Very high	Low

4.3 Key Insight: Cash Flow Inversion

Salaried, student, and retiree archetypes are **retrait-dominant** — money flows out of the account via cash withdrawal. Small business and big business are **versement/virement-dominant** — money flows into or through the account from business operations. This inversion is the strongest archetype discriminator.

5 Population Proportions

Derived from S2 (Amen Bank 2023 deposit mix: 61% individuals / 33% private business / 6% institutional) and Tunisian demographic data (65+ population ∼9%, financial inclusion ∼37% of adults).

Archetype	Share of individuals	Share of total	Calibration source
Salaried	80% of 61%	∼49%	Workforce dominance, mandatory bank accounts
Small business	—	33%	S2 private business segment
Student	8% of 61%	∼5%	Low bank penetration, family dependence
Retiree	12% of 61%	∼7%	65+ demographic (∼9%), high pension-driven ba
Big business	—	6%	S2 institutional segment

Defensibility note: No public dataset provides exact archetype-to-account ratios for Tunisian banks. These proportions are calibrated estimates anchored in demographic data and financial inclusion statistics. The model learns behavioral patterns *within* archetypes, not population ratios — moderate proportion error does not degrade detection capability.

6 Code Architecture

Four classes compose the generator:

Class	Responsibility
`Archetype`	Holds Layer 1 parameters: mean and standard deviation for each of the 6 behavioral dimensions. One instance per archetype (5 total).
`Client`	Holds Layer 2 personality: sampled once at creation from the associated `Archetype` via Gaussian draw. Composition relationship (has-a), not inheritance.
`Event`	Represents a single raw event. Produced by a `Client` with Layer 3 per-event noise applied. Conforms to the locked raw event schema (envelope + operation-specific payload).
`Generator`	Orchestrator. Two-phase execution: (1) planning — creates population, assigns anomaly scenarios and time windows; (2) generation — ticks through time producing events, switching to anomaly logic during assigned windows.

7 Output Format

Design choice	Rationale
Single CSV file	Preserves cross-operation temporal ordering per `client_id`, required for LSTM sequence modeling. Splitting by operation type would break ordering.

Design choice	Rationale
Flat envelope columns + `payload` as JSON string	Envelope fields (`event_id`, `client_id`, `account_id`, `employee_id`, `branch_id`, `timestamp`, `amount`, `currency`, `channel`, `operation_type`) are flat CSV columns. Operation-specific fields are serialized as a JSON string in the `payload` column.
PostgreSQL `COPY` ingestion	The `payload` column maps to a `jsonb` PostgreSQL column. Bulk `COPY` provides fast loading. Generator does not require a running database instance.

Data flow: `Generator` → single CSV → `COPY` → PostgreSQL → batch profile computation → Redis.

8 Anomaly Injection Strategy

8.1 Design Principles

- Training set contains normal data only (both Autoencoder and LSTM learn normal patterns).
- Evaluation set contains normal + labeled anomalies for metric computation (AUC-ROC, F1, Precision, Recall).
- Base anomaly rate: ∼0.13% (anchored from S1 PaySim).
- Two-phase generator: anomaly clients, scenarios, and time windows are assigned during the planning phase before event generation begins.

8.2 Difficulty Gradient

Tier	Share	Characteristics
Easy	10%	Single event, single dimension breaks obviously. Catchable by rule-based systems. Exists because not all fraudsters are sophisticated.
Medium	30%	Single dimension but across multiple events. Requires short-term context (frequency bursts, cumulative thresholds).
Hard	60%	Composite, multi-event, multi-dimension. Each event appears normal in isolation; the anomaly emerges only from the full sequence. This is where ML outperforms rules.

8.3 Scenario Catalog

Tier	Actor	Scenarios
Easy	Client	Amount spike (>5× client average); round suspicious amount (exactly 10,000 DT); duplicate virement (same beneficiary, amount, day); cheque amount inconsistent with history
Easy	Employee	Transaction volume spike vs. peers; error rate spike for experienced employee
Medium	Client	4–5 retraits within 24h; cumulative deposits exceeding threshold within 48h; burst of virements to unknown beneficiaries; operations outside client's normal timing pattern
Medium	Employee	High concentration on specific/new clients; elevated near-threshold ratio; high client-location mismatch ratio
Hard	Client	Smurfing (multiple versements of ∼9,500 DT across days/branches); cheque structuring near 30,000 DT ceiling; money mule pattern (receive-then-forward on new account); gradual behavioral drift over weeks; archetype mismatch (client behaves like wrong archetype)
Hard	Cross-actor	Employee facilitating client fraud — neither actor's metrics are extreme individually, but combined pattern is suspicious (e.g. employee with slightly elevated near-threshold ratio concentrated on a client performing smurfing)

9 Regulatory Update: Law n°41-2024

BCT Q1 2026 tele-compensation data confirms the impact of the August 2024 cheque reform:

- Bank transfers dominate: 9.6M transactions, 65% market share by volume, 0.1% rejection rate.
- Cheques declined 24.9% in volume, 28% in value — historic collapse.
- Promissory notes grew 35.9% in B2B commerce.
- 30,000 DT per-cheque ceiling introduced; banks mandated to encourage alternative payment methods.

Impact on archetypes: remise de chèques reduced across all archetypes, most significantly for big business where virements absorb former cheque volume. New structuring anomaly (D23) added for near-ceiling cheque patterns.